

Longitudinal Power Monitoring in S+C+L Transmission with Forward and Backward Distributed Raman Amplification

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Abstract *We demonstrate longitudinal power monitoring in S+C+L transmission with bidirectional distributed Raman amplification (DRA) for the first time. Measurements over five 80-km spans spatially resolve the effects of DRA and inter-channel stimulated Raman scattering (ISRS) on power evolution, revealing spatial features inaccessible to end-to-end monitoring. ©2026 The Author(s)*

Introduction

Ultra-wideband (UWB) S+C+L transmission with bidirectional distributed Raman amplification (DRA) is a promising approach for achieving >100 Tb/s capacity over long-haul single-mode fibre [1,2]. In such systems, the WDM signal, forward (FW) Raman pump, and backward (BW) Raman pump interact via stimulated Raman scattering, and combined with the wavelength dependence of fibre loss, this produces a complex longitudinal power profile within each span, making system design and optimisation particularly challenging.

Numerical models of such bidirectional Raman-amplified wideband systems have been developed [3–7], but their accuracy relies on precise knowledge of multiple parameters such as wavelength-dependent fibre loss, Raman gain spectra, and effective pump powers. These parameters are difficult to obtain accurately in deployed links. Moreover, end-to-end measurements such as OSNR or Raman on-off gain integrate the combined effects of fibre loss, distributed gain, and inter-channel stimulated Raman scattering (ISRS) over the full span. They cannot resolve how these effects are spatially distributed along the fibre, although such spatial information is essential for pump power allocation and nonlinear noise estimation. A direct measurement of the intra-span power profile would bypass these limitations.

DSP-based longitudinal power monitoring (LPM) estimates the fibre-longitudinal power profile solely from the received coherent signal without additional hardware [8]. Since the foundational demonstrations [9,10], the technique has matured. The linear least squares (LLS) method [11] achieves OTDR-comparable accuracy, and design frameworks for spatial resolution and detectable loss limits have been established [12,13]. LPM has been extended to backward-Raman-amplified C-band links [14], C+L EDFA systems [15], C-band hybrid Raman-EDFA links [16], and fully backward-Raman-amplified C+L links [17]. In our previous work, we demonstrated ISRS

monitoring in S+C+L WDM transmission [18] and visualised forward- and backward-Raman gain evolution in a C-band configuration [19]. To date, however, LPM has been applied only to links with monotonically decreasing power profiles or with Raman gain concentrated in a single region of the span. Bidirectional Raman-amplified S+C+L transmission, where FW gain, BW gain, and ISRS coexist and interact throughout the span, presents a qualitatively different intra-span profile that has yet to be addressed.

In this paper, we present the first LPM demonstration in S+C+L transmission with FW and BW DRA. Since LPM estimates the power profile independently for each WDM channel without requiring any Raman gain or pump parameters, it inherently captures distributed Raman amplification and ISRS-induced inter-band power transfer across all three bands. By comparing profiles measured under FW+BW, FW-only, and BW-only DRA conditions, and under S+C+L and S-band-only transmission, we visualise how FW Raman gain, BW Raman gain, and ISRS jointly shape the intra-span power evolution.

Experimental setup

The experimental setup is shown in Fig. 1(a). The main signal was generated using an offline transmitter (Tx) DSP, a four-channel 120-GSa/s arbitrary waveform generator (AWG), a laser diode (LD), and an IQ modulator (IQM). The modulation format was 100-GBaud PCS-64QAM with a roll-off factor of 0.1. The WDM signals in the S, C, and L bands were emulated using ASE loading with spectral flattening by wavelength-selective switches (WSSs). The main signal and WDM channels were multiplexed on a 150-GHz grid, resulting in a 122-channel 18.3-THz S+C+L-band WDM configuration with 51, 30, and 41 channels in the S, C, and L bands, respectively. The WDM spectrum at the loop input is shown in Fig. 1(c).

A recirculating loop identical to that in [1] was used for the transmission line, comprising an 80-km G.652.D fibre span with bidirectional DRA, an optical repeater with gain equalisers (GEQs), and

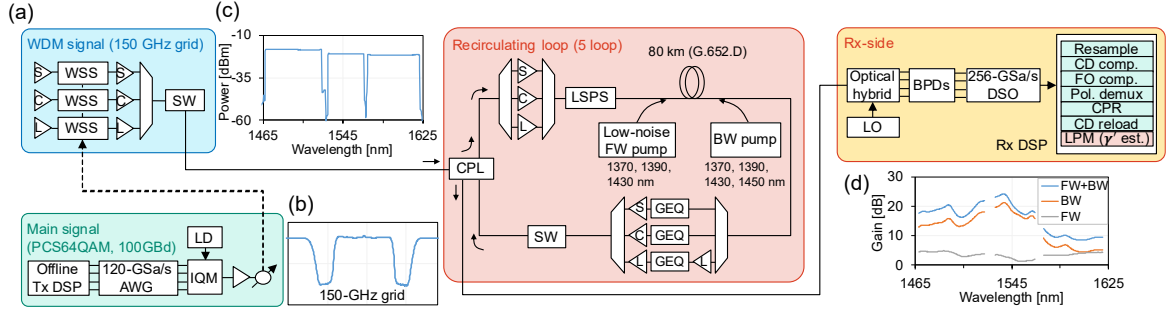


Fig. 1: (a) Experimental setup. (b) 100-Gbaud PCS-64QAM signal spectrum. (c) 18.3-THz S+C+L WDM spectrum at the loop input. (d) Raman on-off gain with low-noise FW and BW pumps.

a loop-synchronous polarisation scrambler (LSPS). The FW low-noise [20] pump wavelengths and powers were 1370, 1390, and 1430 nm, and 201.4, 210, and 145 mW, respectively. The BW pump wavelengths and powers were 1370, 1390, 1430, and 1450 nm, and 306, 455, 185, and 50 mW, respectively. The resulting Raman on-off gain is shown in Fig. 1(d). The signal was transmitted over 5 loops (400 km). The total fibre input powers in the S, C, and L bands were 16.64, 12.75, and 14.34 dBm, respectively.

On the Rx side, the main signal was demultiplexed per band and detected by a coherent receiver with a 256-GSa/s digital storage oscilloscope (DSO). The offline Rx DSP chain consisted of resampling, chromatic dispersion (CD) compensation, frequency offset compensation, polarisation demultiplexing, carrier phase recovery (CPR), and CD reloading. The power profile $P(z)$ was then estimated by the linear least squares (LLS) method [11], which obtains the effective nonlinear coefficient vector $\gamma' \equiv \gamma P$ as $\hat{\gamma}' = (\text{Re}[\mathbf{G}^\dagger \mathbf{G}])^{-1} \text{Re}[\mathbf{G}^\dagger \mathbf{A}_1]$. Since γ is known, $P(z)$ is directly obtained.

To investigate how FW Raman gain, BW Raman gain, and ISRS jointly shape the longitudinal power evolution, LPM was measured under four conditions: (i) S+C+L transmission with FW+BW DRA, (ii) S+C+L with FW DRA only, (iii) S+C+L with BW DRA only, and (iv) S-band-only with FW+BW DRA. When active, each pump was operated at the wavelengths and powers listed above. Only the GEQ settings and repeater

amplifier gains were adjusted between conditions. LPM was performed at the centre wavelength of each band: 1494.67 nm (S), 1547.12 nm (C), and 1595.28 nm (L). For each wavelength, 20 independent data sequences were each captured 5 times, and the resulting 100 power profiles were averaged. A three-point moving average was then applied. The step size was $\Delta z = 1$ km, and the wavelength dependence of γ was corrected following Eq. (4) in [7].

Results and discussion

Fig. 2(a) shows the measured longitudinal power profiles of representative S-, C-, and L-band channels over five 80-km spans under S+C+L transmission with bidirectional DRA. All three bands exhibit non-monotonic profiles shaped by the combined effects of fibre loss, FW and BW Raman gain, and ISRS-induced inter-band power transfer. At the repeater sections, the L band exhibits a sharp power step due to a lumped EDFA before the GEQ. The dip-like features in the S and C bands at these locations are attributed to GEQ loss. To quantify the spatially varying contributions of these mechanisms, we calculated the local slope of the LPM-measured power profiles using a smoothed second-order polynomial fit, as shown in Fig. 2(b) for the fifth span. We focus on the 10–70 km range, where the slope reflects fibre-intrinsic physics. The 0–10 km and 70–80 km ranges are affected by boundary effects and repeater components such as the L-band EDFA before the GEQ. In the first half, all three bands show slopes around 0 to -0.17

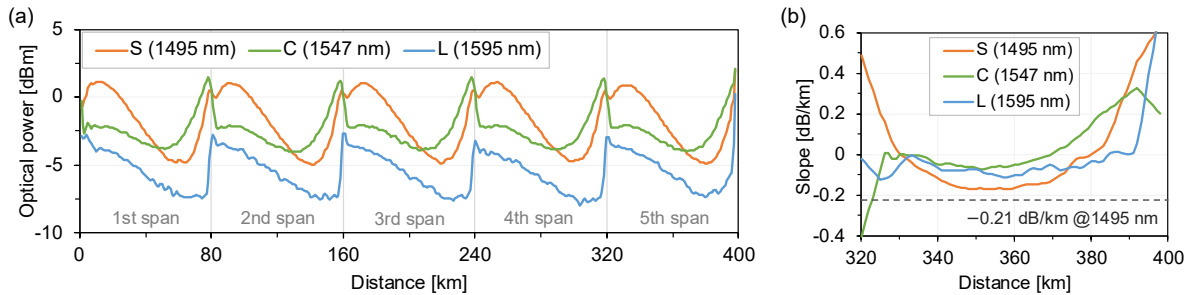


Fig. 2: (a) Measured longitudinal power profiles of S-, C-, and L-band channels over five 80-km spans with bidirectional DRA. (b) Local slopes for the fifth span.

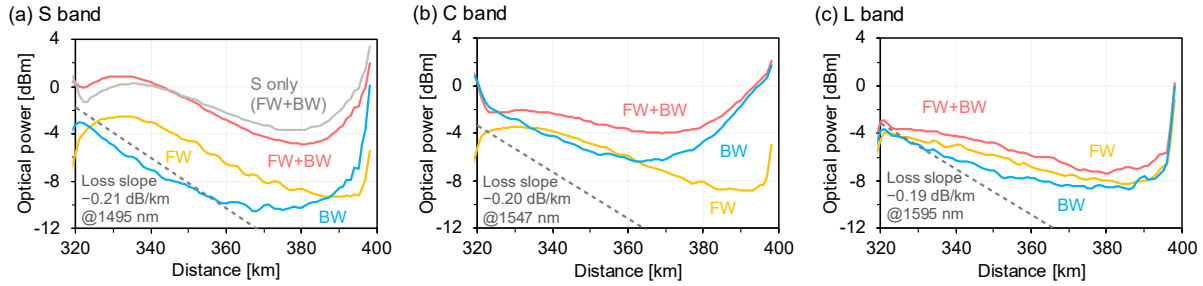


Fig. 3: Measured longitudinal power profiles for the fifth 80-km span in the (a) S, (b) C, and (c) L bands, under four conditions: S+C+L transmission with FW+BW, FW-only, and BW-only DRA, and S-band-only transmission with FW+BW DRA.

dB/km, well above the intrinsic-fibre-loss slopes of -0.21 dB/km (S) and -0.19 dB/km (L). This is due to FW Raman gain in the S and C bands, ISRS-induced power transfer from shorter wavelengths in the C and L bands, and reduced intrinsic loss in the L band. In the second half, BW Raman amplification manifests differently across bands. The S-band slope reverses sharply beyond ~ 60 km, reaching $\sim +0.3$ dB/km at 70 km. The C-band slope reverses more gradually. The L-band slope stays around -0.1 dB/km, still above -0.19 dB/km, indicating that ISRS-induced power transfer together with modest direct gain from the 1450-nm BW pump offsets the fibre loss throughout the span. These spatially resolved slopes reveal how FW Raman gain, BW Raman gain, and ISRS contribute differently to the first and second halves of the span.

LPM was further measured under FW-only DRA, BW-only DRA, and S-band-only transmission with FW+BW DRA. Fig. 3 shows the resulting power profiles for the fifth 80-km span in the S, C, and L bands. The contrast between the S and L bands is clearest, reflecting the different interplay of FW Raman gain and ISRS in each band. In the S band, FW Raman gain provides direct amplification. Under FW-only DRA, the S- and C-band power is gently lifted in the first half with a subsequent slope gentler than the intrinsic fibre loss, indicating that FW pump gain persists throughout the span. Under BW-only DRA, the absence of this direct gain leaves ISRS-induced power transfer to the C and L bands uncompensated, and the S-band power drops steeply in the first half. BW Raman amplification then lifts the S band sharply in the second half, producing a spatial profile markedly different from that of FW-only DRA even though the end-of-span powers are similar. In the L band, FW Raman gain acts indirectly through SRS. Under FW-only DRA, the L-band power is lifted in the first half despite receiving no direct FW pumping, illustrating that FW-amplified S-band power is redistributed to the L band along the fibre via ISRS. Under BW-only DRA, this power transfer is reduced because the S-band is depleted, and the L band attenuates more rapidly. In the second half, BW Raman gain

provides weak direct amplification to the L band from the 1450-nm pump. The C band shows intermediate behaviour, with weak direct gain from the 1430-nm FW pump and ISRS-induced power transfer from the S band under FW-only DRA, and a steepened first-half slope under BW-only DRA. Comparison between S+C+L and S-band-only transmission under FW+BW DRA further highlights the ISRS contribution. The S-band-only case shows a noticeably gentler first-half slope, consistent with the absence of ISRS-induced power transfer to the C and L bands. The increasing difference with distance indicates that ISRS-induced power transfer accumulates along the fibre. These observations directly visualise the spatially complementary roles of FW and BW Raman gain and the ISRS-induced inter-band coupling across all three bands.

Conclusions

We demonstrated, for the first time, DSP-based LPM in S+C+L transmission with bidirectional DRA. The non-monotonic power evolution shaped by the interplay of FW gain, BW gain, and ISRS was captured across all three bands without requiring any Raman gain coefficients or pump parameters. Measurements under FW-only, BW-only, and S-band-only conditions further revealed the spatially complementary roles of FW Raman gain, BW Raman gain, and ISRS-induced inter-band power transfer. These results demonstrate that LPM can visualise the complex longitudinal power evolution of bidirectionally Raman-amplified UWB systems, offering a potential complement to existing design methodologies and opening a path toward more advanced system design through combination with them.

Acknowledgements

This work was supported in part by the commissioned research (No. JPJ012368G60201) of the National Institute of Information and Communications Technology (NICT), Japan.

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